

Spring 1983

Problem 1 (Sp83). Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be a monotone decreasing function, defined on the positive real numbers with

$$\int_0^\infty f(x) dx < \infty$$

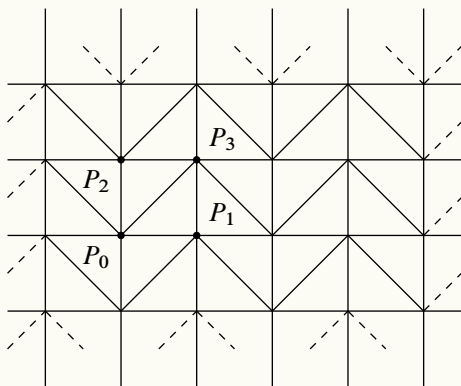
Show that

$$\lim_{x \rightarrow \infty} xf(x) = 0$$

Problem 2 (Sp83, Fa11, Sp21). Let A be an $n \times n$ real matrix where all entries not on the diagonal are negative, and the sum of the entries in each column is positive. Show that the determinant of A is positive.

Problem 3 (Sp83). A fractional linear transformation maps the annulus $r < |z| < 1$ with $r > 0$, onto the domain bounded by the two circles $|z - 1/4| = 1/4$ and $|z| = 1$. Find r .

Problem 4 (Sp83). In the triangular network in $\mathbb{R}^2$ depicted below, the points P_0, P_1, P_2 , and P_3 are respectively $(0, 0)$, $(1, 0)$, $(0, 1)$, and $(1, 1)$. Describe the structure of the group of all Euclidean transformations of $\mathbb{R}^2$ which leave this network invariant.



Problem 5 (Sp83). Find all solutions $y : \mathbb{R} \rightarrow \mathbb{R}$ to

$$\frac{dy}{dx} = \sqrt{y(y-2)} \quad y(0) = 0$$

Problem 6 (Sp83, Sp01, Fa19). Suppose that f is a continuous function on $\mathbb{R}$ which is periodic with period 1. Show that:

1. The function f is bounded above and below and achieves its maximum and minimum.
2. The function f is uniformly continuous on $\mathbb{R}$.
3. There exists a real number x_0 such that $f(x_0 + \pi) = f(x_0)$.

Problem 7 (Sp83). Let $\mathbb{Z}_p$ be the group of integers mod p , under addition, where p is a prime number. Suppose that n is an integer satisfying $1 \leq n \leq p$, show that the group $\mathbb{Z}_p \times \mathbb{Z}_p \times \cdots \times \mathbb{Z}_p$, for a total of n factors, has no automorphism of order p^2 .

Problem 8 (Sp83). Suppose that $n > 1$ is an integer. Prove that the sum

$$1 + \frac{1}{2} + \cdots + \frac{1}{n}$$

is not an integer.

Problem 9 (Sp83). Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous and satisfies

$$\|f(x) - f(y)\| \geq \lambda \|x - y\|$$

for all $x, y \in \mathbb{R}^n$ and some $\lambda > 0$. Prove that f is one-to-one, onto, and has a continuous inverse.

Note: See also ??.

Problem 10 (Sp78, Sp83, Sp97). Evaluate $\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx$.

Problem 11 (Sp83, Fa21, Sp20, Fa20). Let A be an invertible real $n \times n$ matrix. Show that there is a decomposition $A = QR$ in which Q is an $n \times n$ real orthogonal matrix and R is upper triangular with positive diagonal entries. Is this decomposition unique?

Problem 12 (Sp83, Sp18). Determine all the complex analytic functions f defined on the unit disc $\mathbb{D}$ which satisfy

$$f''\left(\frac{1}{n}\right) + f\left(\frac{1}{n}\right) = 0$$

for $n = 2, 3, 4, \dots$

Problem 13 (Sp83, Sp10, Fa24). Let $\lambda_1, \lambda_2, \dots, \lambda_n, \dots$ be real numbers. Show that the infinite series

$$\sum_{n=1}^{\infty} \frac{e^{i\lambda_n x}}{n^2}$$

converges uniformly over $\mathbb{R}$ to a continuous limit function $f : \mathbb{R} \rightarrow \mathbb{C}$. Show, further, that the limit

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T f(x) \, dx$$

exists.

Problem 14 (Sp83). Let G be an abelian group which is generated by, at most, n elements. Show that each subgroup of G is again generated by, at most, n elements.

Problem 15 (Sp79, Sp83). Let P and Q be complex polynomials with the degree of Q at least two more than the degree of P . Prove there is an $r > 0$ such that if C is a closed curve outside $|z| = r$, then

$$\int_C \frac{P(z)}{Q(z)} \, dz = 0$$

Problem 16 (Sp83). Let V be a real vector space of dimension n , and let $S : V \times V \rightarrow \mathbb{R}$ be a nondegenerate bilinear form. Suppose that W is a linear subspace of V such that the restriction of S to $W \times W$ is identically 0. Show that $\dim W \leq n/2$.

Problem 17 (Sp83). Evaluate $\int_{-\infty}^{\infty} \frac{\sin x}{x(x - \pi)} \, dx$.

Problem 18 (Su79, Sp82, Sp83, Su84, Fa91, Fa98, Sp11, Fa15, Fa16, Fa23). Let $\mathbb{F}$ be a finite field with q elements and let V be an n -dimensional vector space over $\mathbb{F}$.

1. Determine the number of elements in V .
2. Let $\text{GL}_n(\mathbb{F})$ denote the group of all $n \times n$ nonsingular matrices over $\mathbb{F}$. Determine the order of $\text{GL}_n(\mathbb{F})$.
3. Let $\text{SL}_n(\mathbb{F})$ denote the subgroup of $\text{GL}_n(\mathbb{F})$ consisting of matrices with determinant 1. Find the order of $\text{SL}_n(\mathbb{F})$.

Problem 19 (Su77, Su80, Sp82, Sp83). Prove that the initial value problem

$$\frac{dx}{dt} = 3x + 85 \cos x \quad x(0) = 77$$

has a solution $x(t)$ defined for all $t \in \mathbb{R}$.

Problem 20 (Sp83). Show that the interval $[0, 1]$ cannot be written as a countably infinite disjoint union of closed subintervals of $[0, 1]$.